Tiankui Zhang

Geometric Kernel & CAD Algorithm Engineer · PhD in Physics

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Education

University of Arizona
Tucson, AZ, USA

PhD in Physics (Computational Physics), GPA: 4.0/4.0 — *Aug 2014 - Jul 2020*

- Advisor: Prof. Charles Wolgemuth

Wuhan University
Wuhan, China

Bachelor of Science in Physics, GPA: 3.85/4.0 — *Sep 2010 - Jun 2014*

- Top-Notch Students Scientific Development Program (Physics)
- Advisor: Dr. Eugene Lim (King's College London)

King's College London
London, UK

International Student Exchange Program — *Sep 2013 - May 2014*

Experience

Poissonsoft
Shanghai, China

Software Engineer, Geometric Kernel — *Feb 2025 - Present*

- **Procedural geometry performance**: optimized the evaluation of procedural intersection curves and rolling-ball blend surfaces, each evaluation of which requires iterative solving; profiled bottlenecks and improved solving strategy, initial guesses, derivative computation and reuse of intermediate results, without sacrificing accuracy or robustness.
- **Local operations and topology editing**: maintained and developed B-rep modeling APIs for face taper, body taper, redundant topology removal and sewing, keeping geometry and topology consistent under degenerate cases and tolerances.

Xindi Digital
Shanghai, China

Senior Researcher, CAD Features — *Jun 2023 - Jan 2025*

- **Offset curve**: developed end to end, from UI to algorithm; offsets chains of planar curves of any type by distance or draft angle, handling local and global self-intersections, corner gaps and vanishing segments, and succeeds where the kernel's built-in offset fails.
- **Curve fitting**: developed end to end; approximates curves with B-splines of a given degree and segment count under endpoint continuity constraints up to G2, formulated as a constrained quadratic program.
- **Surface fairing**: developed end to end; recomputes B-spline control points to minimize second- and third-order derivative energy while preserving boundary continuity.
- **Boundary blend**: a comprehensive surface-generation feature: lofting from curves in one direction, Coons and Gordon surfaces from two directions, with continuity constraints to adjacent surfaces.
- **Bridge surface**: smooth transition surface between edges of two surfaces, with specified continuity, tangent magnitude and flow.
- **Draft offset**: offsets a sketch-bounded region of a surface by a given distance, with side walls at a given draft angle.

Glodon
Shanghai, China

Geometric Algorithm Engineer — *Sep 2020 - Jun 2023*

- **Adaptive watertight tessellation**: rewrote B-rep tessellation for display and downstream computation; edges are discretized first and shared by adjacent faces to guarantee watertightness, and faces are adaptively refined in parameter space until the 3D error meets tolerance. Triangle count dropped substantially, on par with ACIS.
- **Coplanar edge hiding**: splits edges where coplanar faces of two bodies overlap, so the renderer can hide the seam between joined components.
- **MEP occlusion in 2D drawings**: computes the occluded portions of pipes and equipment from the 3D model and draws them broken or dashed per drafting conventions.
- **Silhouette curves**: computes silhouette curves of surfaces for a given view direction.
- **2D geometry comparison**: similarity measure between 2D shapes for AI applications under a large distance tolerance, based on each shape's region of influence.

University of Arizona *Tucson, AZ, USA*

Research Assistant and Teaching Assistant *Aug 2014 - May 2020*

- **Sixth-order level set schemes**: designed sixth-order accurate schemes for reinitialization and extrapolation in the level set framework, building on WENO schemes for Hamilton–Jacobi equations (*J. Sci. Comput.*, 2020).
- **Membrane dynamics**: derived restoring forces from a generalized curvature energy using differential geometry and variational calculus, and built a stable level-set framework for large deformations of single- and multi-phase vesicles, including fission and protein–membrane interaction (*J. Comput. Phys.*, 2022).
- **GPU implementation**: implemented the 3D framework in MATLAB and CUDA C++, coupling the level set equation with diffusion–convection equations on the evolving surface.
- **Teaching**: teaching assistant for undergraduate physics and astronomy courses.

Skills

Programming	C++, CUDA, MATLAB
Geometric Modeling	B-rep solid modeling and topology editing; NURBS curves and surfaces; intersection and blending; tessellation and mesh generation
Numerical Methods	constrained optimization (quadratic programming); level set methods; finite volume; high-order schemes for Hamilton–Jacobi equations
Theory	differential geometry, variational calculus, physics
Languages	Chinese (native), English (full professional proficiency)

Publication

- **Tiankui Zhang** and Charles W Wolgemuth. Sixth-order accurate schemes for reinitialization and extrapolation in the level set framework. *Journal of Scientific Computing*, 83(2), 2020.

- **Tiankui Zhang** and Charles W Wolgemuth. A general computational framework for the dynamics of single- and multi-phase vesicles and membranes. *Journal of Computational Physics*, Volume 450, 2022, 110815, ISSN 0021-9991.